

$$\begin{array}{ccccc}
F^T[X, Y] & \xrightarrow{F^T \Gamma_{X, Y}^Z} & F^T[[Z, X], [Z, Y]] & \xrightarrow{\psi_{[Z, X], [Z, Y]}} & [F^T[Z, X], F^T[Z, Y]]_T \\
\psi_{X, Y} \downarrow & & & & \downarrow [\text{id}, \psi_{Z, Y}]_T \\
[F^T X, F^T Y]_T & \xrightarrow{\Gamma_{F^T X, F^T Y}^{F^T Z}} & [[F^T Z, F^T X]_T, [F^T Z, F^T Y]_T]_T & \xrightarrow{[\psi_{Z, X}, \text{id}]_T} & [F^T[Z, X], [F^T Z, F^T Y]_T]_T
\end{array}$$